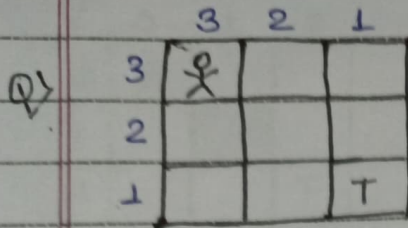


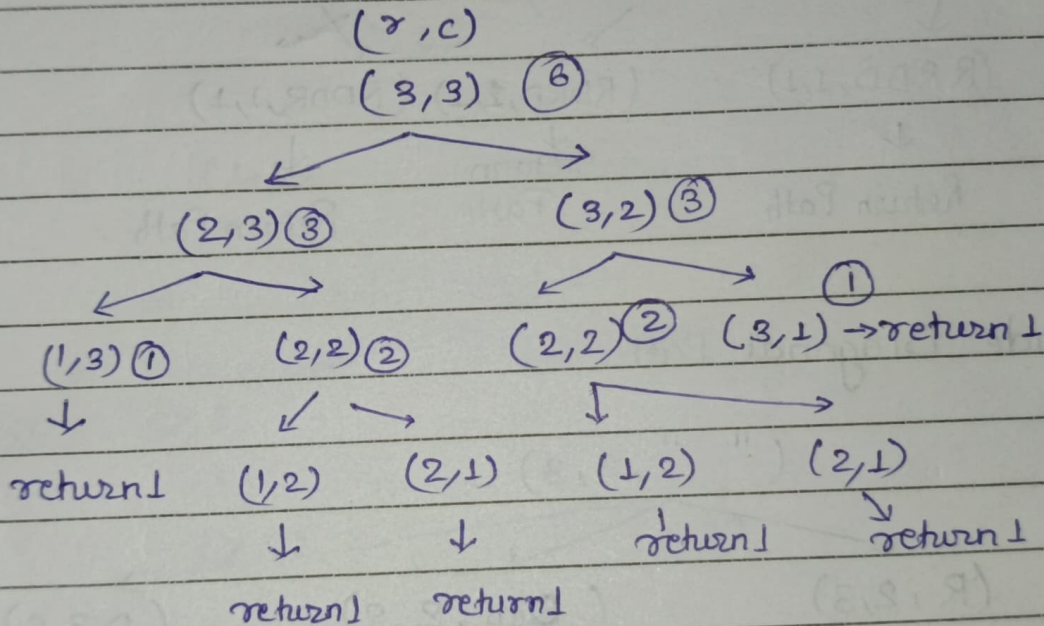
BackTracking + Maze Problems



⊗ → R
↓ D.

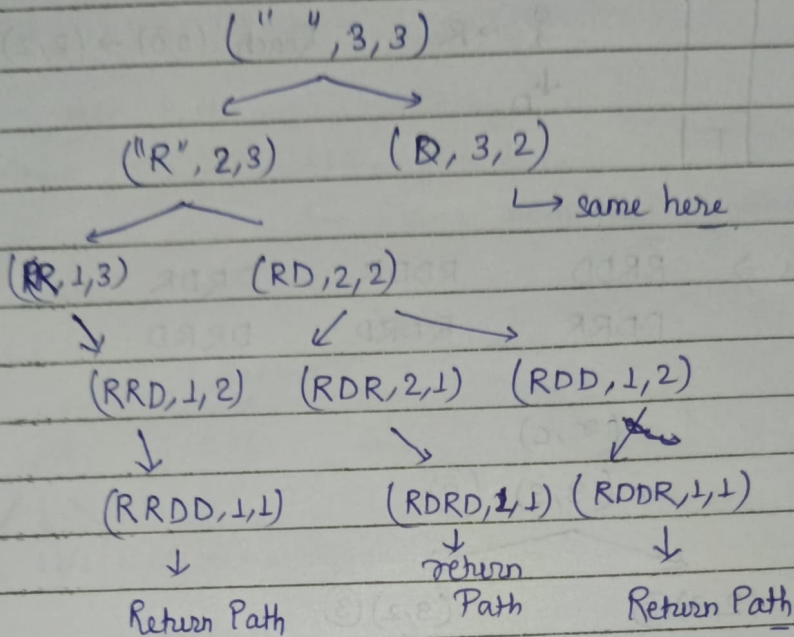
Reach (0,0) → (2,2)

Ans ⇒ RRDD RDDR DRDR
 DDDR RDRD DRRD



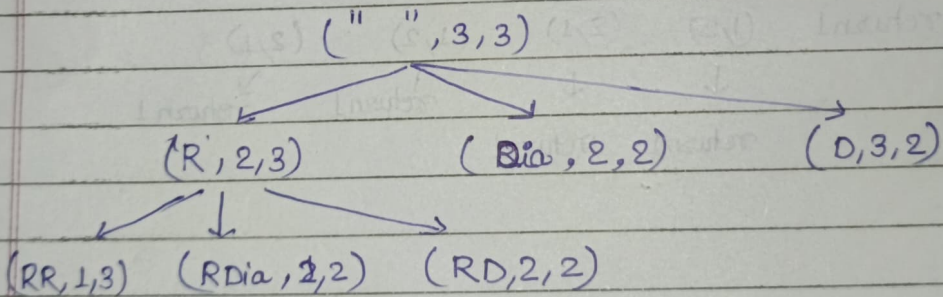
Q27

Displaying Paths



Q37

Include Diagonal Path



4) Maze with obstacle

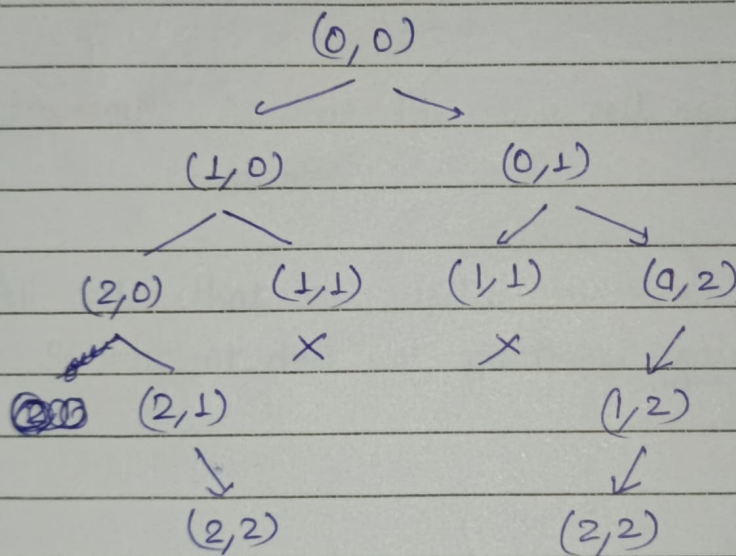
	0	1	2
0	⊗		
1		■	
2			F



— Obstacle

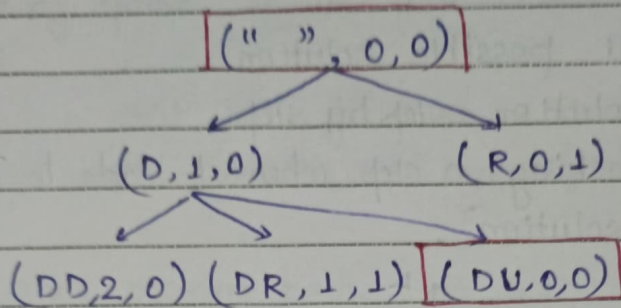
Note:- when you land on a new cell check whether it is an obstacle or not.

If you land on recursion stop the recursion.



Q>

	0	1	2
0	X		
1			
2			F



Problem:- It visit the same cell again so recursion will not stop.

- All cells that are visited mark it as false.
So that it does not go there again.

Q. What is Backtracking?

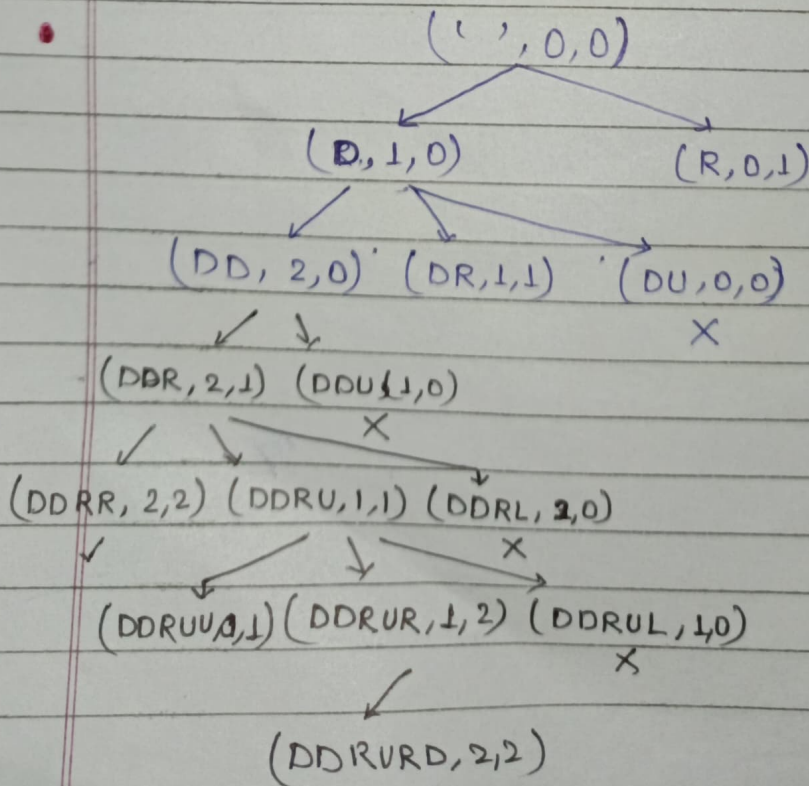
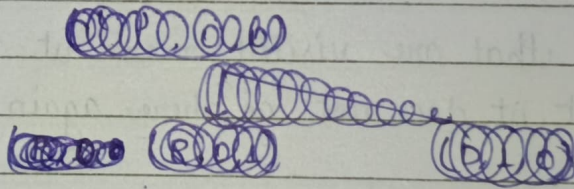
Backtracking is a general algorithm technique that involves :-

- Trying out all possible solution
- Building the solution step by step.
- Undoing (backtracking) a step when it leads to a dead end (invalid solution).

	0	1	2
0	0 F	.	
1	F	F	
2	F	F	Final

A stick figure is shown with four arrows pointing outwards from its center, representing forces. The arrows are labeled: 'L' for left, 'R' for right, 'U' for up, and 'D' for down.

- marking false.



Problem:- Original array is modified
means for future recursion it gets modified

- Marking false means that cell is present in current path.
- So, when that path is over, another recursion should get the actual maze, not modified one.
- So, while moving back, it ~~is~~ maze should be restored like original maze.

Backtracking:- It means make a change & ~~remove~~
remove it when work is done

Q64 Print matrix of Path.

1		
2		
3	4	5

DDRR

1		
2	5	6
3	4	7

DDRURD

- Take a step variable
- Update the path array
- Print it in base condition
- Backtrack.